MOHAMMADHOSSEIN MOHAMMADNIA

Solid-Earth Geodesy · InSAR Time Series · Volcano Deformation & Hazard Assessment

Ph.D. thesis defended and passed on 23 July 2026 · PhD award letter not yet received — expected by September 2026

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RESEARCH PROFILE

Solid-Earth geodesist specialising in Interferometric Synthetic Aperture Radar (InSAR) for volcano and tectonic deformation monitoring. My work **develops new time-series and atmospheric-correction methods** — including the open-source **Common-Mode Filtering (CMF)** algorithm — and applies them to detect subtle unrest at remote, unmonitored volcanic systems. First-authored publications in *Geophysical Research Letters*, *Geophysical Journal International*, *Bulletin of Volcanology* and *Tectonophysics* report the first documented volcanic unrest at Taftan volcano (Makran arc) and the first province-wide deformation baseline for the Tibesti Volcanic Province. I integrate InSAR, GNSS, pixel-offset tracking, Bayesian geodetic inversion and petrological–thermodynamic modelling to connect surface deformation to subsurface magmatic and hydrothermal processes, and I am extending this toolkit with machine-learning approaches towards **automated, operational volcano monitoring**.

RESEARCH INTERESTS

- **Geodetic remote sensing:** multi-temporal InSAR (SBAS, PS/DS, phase-bias mitigation), pixel-offset tracking and optical flow, InSAR–GNSS integration, and atmospheric noise correction for millimetre-scale deformation.
- **Volcano geodesy & hazard:** detection and physical interpretation of unrest at long-dormant and poorly instrumented volcanoes; magmatic vs. hydrothermal source discrimination; eruption-cycle deformation; volcano risk reduction in data-scarce regions.
- **Geophysical inversion & modelling:** Bayesian (MCMC) source inversion, analytical and numerical elastic/viscoelastic models, thermodynamic (Rhyolite-MELTS) and thermobarometric constraints on magma storage.
- **Solid-Earth deformation more broadly:** salt tectonics and diapirism, land subsidence and aquifer systems, glacier dynamics and cryosphere change.
- **Computational geoscience:** machine/deep learning for automated deformation detection, large-scale SAR data pipelines and cloud/HPC processing workflows.

EDUCATION

Ph.D. in Earth Sciences · The University of Hong Kong **2022 – 2026**

Hong Kong SAR, China

- **Dissertation:** ***[Deformation survey of remote and poorly monitored volcanoes with operational satellite radar interferometry missions]***
- Supervisors: Dr. Pablo J. González (IPNA-CSIC, Spain) and Prof. Dr. A. Alexander G. Webb (Freie Universität Berlin / HKU).
- **Degree status:** oral examination (viva) successfully passed on 23 July 2026; the thesis was approved by the examination panel subject to minor amendments. **The PhD award letter has not yet been received.** Under University of Hong Kong regulations, the Graduate School issues the Letter of Award after submission of the final amended thesis; issuance is expected by September 2026.
- Awarded the HKU Postgraduate Scholarship (PGS), 2022–2026.

M.Sc. in Geophysics · Institute for Advanced Studies in Basic Sciences (IASBS) **2016 – 2019**

Zanjan, Iran

- **Thesis:** ***[Surface deformation mapping of salt structures in the Kalut Basin in Central Iran, using InSAR method]*** (Supervisor: Dr. Zahra Mousavi – Dr Mahdi Najafi).
- Graduated **first in class** — Academic Excellence Award. GPA: **[18.13/20]**

B.Sc. in Physics · Azarbaijan Shahid Madani University **2011 – 2014**

Tabriz, Iran

RESEARCH APPOINTMENTS & EXPERIENCE

Research Associate · Department of Earth Sciences, The University of Hong Kong **02/2026 – 05/2026**

Hong Kong SAR, China

– Developing an efficient geocoded-SLC processing workflow for operational, large-scale satellite volcano deformation monitoring (manuscript in preparation).

Doctoral Researcher · The University of Hong Kong / IPNA-CSIC **02/2022 – 02/2026**

Hong Kong SAR, China · Tenerife, Spain

- Designed and released the **InSAR Common-Mode Filtering (CMF)** algorithm — a data-driven atmospheric-noise mitigation method that reduces time-series standard deviation by ~36% on its own and by up to ~50% when combined with ERA5 corrections; published in *Geophysical Research Letters* and openly released via GitHub and Zenodo.
- Detected and characterised the **first documented volcanic unrest at Taftan volcano** (Makran subduction arc): a trigger-less, 10-month, ~9 cm summit uplift peaking at 11 cm/yr, sourced at 490–630 m depth; constrained onset and cessation using hyperbolic-tangent transient modelling and Bayesian (GBIS) source inversion.
- Produced the **first province-wide InSAR deformation baseline for the Tibesti Volcanic Province** (Chad, ~29,000 km²), processing Sentinel-1 data over three descending frames across eight volcanic centres; identified persistent subsidence at Toussidé–Yirrigué (−2.2 ± 0.3 mm/yr, 7.6 km source) and Soborom (−8.5 ± 0.3 mm/yr, 4.9 km source).
- Integrated geodesy with **petrological thermobarometry and Rhyolite-MELTS thermodynamic modelling**, plus viscoelastic cooling models, to attribute Tibesti subsidence to Holocene-emplaced, cooling and degassing magma reservoirs — a rare cross-disciplinary geodetic–petrological synthesis.
- Diagnosed moisture-driven “fading-signal” phase-bias artefacts over saline caldera floors and demonstrated a long-temporal-baseline stacking test to distinguish artefacts from real deformation — a methodological caution now applicable to InSAR studies across arid, evaporitic terrains.
- Research conducted within the Spanish AEI *VolcaMotion* project and in association with the ERC Synergy Grant *ROTTnROCK*; contributed to atmospheric-correction work at Teide–Pico Viejo, Tenerife (2015–2025).

Visiting Researcher · Stockholm University **09/2019 – 12/2019**

Stockholm, Sweden — funded by the Bolin Centre for Climate Research (Arctic Avenue programme)

- Applied time-series D-InSAR/SBAS to 135 Sentinel-1A/B acquisitions (2017–2019, ascending + descending, VV/VH) to quantify mass balance and flow of five glaciers in the Tarfala–Kebnekaise region, northern Sweden; resolved up to 1.2 m ice loss and maximum surface velocities of 2.3 m/yr.

Graduate Researcher · Institute for Advanced Studies in Basic Sciences (IASBS) **2016 – 2019**

Zanjan, Iran

- Processed 137 Sentinel-1 SLC scenes and generated 724 interferograms (GMTSAR + SNAPHU, SBAS) to deliver the **first basin-wide kinematic survey of an entire diapiric province**, decomposing ascending/descending LOS velocities into vertical and horizontal fields.
- Established the Kalut basin (Central Iran) as one of the most actively rising exposed diapiric systems worldwide (uplift to 8 mm/yr), and classified salt walls, stocks, massifs and welds by their distinct kinematic signatures.

PEER-REVIEWED JOURNAL PUBLICATIONS

*Author name in bold. * denotes corresponding author. Citation metrics:*

1. **Mohammadnia, M.**, Boulesteix, T., Webb, A. A. G., & González, P. J. (2026). Satellite surface deformation survey reveals evidence for active magmatic processes at the Tibesti (Africa) continental intraplate volcanoes. *Bulletin of Volcanology*, 88, 37. https://doi.org/10.1007/s00445-026-01954-0

   First satellite geodetic evidence of active magmatic processes in an African continental intraplate volcanic province.

2. **Mohammadnia, M.**, Yip, M. W., Webb, A. A. G., & González, P. J. (2025). Spontaneous transient summit uplift at Taftan volcano (Makran subduction arc) imaged using an InSAR common-mode filtering method. *Geophysical Research Letters*, 52(19), e2025GL114853. https://doi.org/10.1029/2025GL114853

   Widely covered in international media — see ***Research Impact & Media Coverage****.*

3. **Mohammadnia, M.***, Vajedian, S., Mousavi, Z., & Aflaki, M. (2025). Surface deformation dynamics of Sierra Negra's 2018 eruption: insights from InSAR, optical flow and pixel offset tracking. *Geophysical Journal International*, 240(3), 1773–1789. https://doi.org/10.1093/gji/ggaf012

   First estimate of subsurface magma transfer rate (~60 m/hr) from a one-day ascending–descending SAR offset.

4. **Mohammadnia, M.**, Najafi, M., & Mousavi, Z. (2021). InSAR constraints on the active deformation of salt diapirs in the Kalut basin, Central Iran. *Tectonophysics*, 810, 228860. https://doi.org/10.1016/j.tecto.2021.228860

   One of the first comprehensive InSAR studies of an entire diapiric basin.

MANUSCRIPTS UNDER REVIEW & IN PREPARATION

1. Ajorlou, N., **Mohammadnia, M.**, Aflaki, M.*, Mousavi, Z.*, & Vajedian, S. A comprehensive satellite-based survey of salt diapir deformation and classification in the Fars arc of Zagros, Iran. *Tectonophysics* — **under review**.

2. González, P. J., Boulesteix, T., **Mohammadnia, M.**, Webb, A. A. G., & Charco, M. CO_2 flushing drives bubble nucleation and pressurization in phonolitic magma reservoirs. **In preparation**.

3. **Mohammadnia, M.**, & González, P. J. Efficient satellite volcano deformation monitoring using a geocoded SLC workflow. **In preparation**.

SOFTWARE, DATA & OPEN SCIENCE

- **Common-Mode Filtering (CMF) for InSAR time series** — lead developer. Open-source, peer-reviewed atmospheric-noise mitigation tool for small-area volcano deformation; designed for integration into MintPy, StaMPS and LiCSBAS workflows. GitHub · Zenodo: 10.5281/zenodo.14629139
- Adopted in subsequent published research (Mohammadnia et al., *Bulletin of Volcanology*, 2026) as the standard filtering step for volcanic-centre time series.
- Committed to reproducible, open science: all publications are open access; derived deformation products and processing scripts released alongside papers.

RESEARCH IMPACT & MEDIA COVERAGE

The Taftan volcano study (GRL, 2025) received global press attention in October 2025, reaching general audiences across Europe, North America and the Middle East:

- Live Science · The Independent · New York Post · Daily Mail · Yahoo News
- Earth.com · ZME Science · The Watchers · The Brighter Side of News · Illustreret Videnskab (Denmark) · bulletin (UK)
- The study's central recommendation — that permanent volcano monitoring networks are needed along the Makran arc was carried by outlets in several countries, translating a methodological advance into a concrete public-safety message.

CONFERENCE CONTRIBUTIONS & INVITED TALKS

1. González, P. J., **Mohammadnia, M.**, Boulesteix, T., Webb, A. A. G., & Charco, M. (2026). Atmospherically corrected deformation at a prominent-topography volcano: Teide–Pico Viejo, Tenerife, Canary Islands (2015–2025). *EGU General Assembly 2026*, Vienna, Austria. (**accepted**)

2. González, P. J., Boulesteix, T., & **Mohammadnia, M.** (2025). Teide volcano: review of its eruptive history and current active hydrothermal system. *1st ERC Synergy Grant ROTTnROCK Assembly*, Uppsala, Sweden, May 2025.

3. **Mohammadnia, M.**, Yip, M. W., Webb, A. A. G., & González, P. J. (2025). Spontaneous transient summit uplift at Taftan volcano (Makran subduction arc) imaged using an InSAR common-mode filtering method. *EGU General Assembly 2025*, Vienna, Austria.

4. **Mohammadnia, M.**, Mousavi, Z., & Najafi, M. (2021). InSAR constraints on active deformation of salt structures in the Kalut basin, Central Iran. *FRINGE 2021*, Delft, The Netherlands.

5. Darvishi, M., **Mohammadnia, M.**, & Jaramillo, F. (2021). Monitoring glacier dynamics in Tarfala Valley, Sweden with Sentinel-1 data. *FRINGE 2021*, Delft, The Netherlands.

6. Shahbazi, S., Mousavi, Z., Rezaei, A., & **Mohammadnia, M.** (2021). Characterization of hydrological properties through joint InSAR–well measurements, Salmas Plain, NW Iran. *FRINGE 2021*, Delft, The Netherlands.

7. Jaramillo, F., Aminjafari, S., **Mohammadnia, M.**, Darvishi, M., & García, J. (2020). Using InSAR observations to understand changes to coastal and inland water resources in the Baltic basin. *3rd Baltic Earth Conference*, Jastarnia, Hel Peninsula, Poland, p. 199.

HONOURS, AWARDS & FUNDING

Ph.D. Thesis Evaluation — Rated "Excellent" (Top 10%) by examiners, The University of Hong Kong — July 2026

HKU Postgraduate Scholarship (PGS) — The University of Hong Kong — 2022 – 2026
Competitive full doctoral scholarship covering stipend and tuition.

Academic Excellence Award — ranked 1st in M.Sc. cohort, IASBS — 2019

Bolin Centre for Climate Research / Arctic Avenue research funding — Stockholm University — 2019
Funded visiting-researcher position for Arctic glacier and Baltic hydro-geodesy research.

TECHNICAL SKILLS

- **InSAR & SAR processing:** ISCE2, GMTSAR, MintPy, StaMPS, LiCSAR / LiCSBAS, MiaplPy, SNAP, SARscape, DORIS, NSBAS, AMSTer, Dolphin, COMPASS, HyP3 (cloud), SNAPHU, GAMMA-based workflows.

- **Modelling & inversion:** GBIS (Bayesian MCMC source inversion), Mogi / McTigue / Yang / Okada analytical sources, elastic and Maxwell-viscoelastic deformation modelling, Rhyolite-MELTS thermodynamic modelling, DefVolc, Pyrocko.
- **Programming & data science:** Python (NumPy, SciPy, GDAL, OpenCV, Matplotlib), MATLAB, Bash / Linux shell, Git & version control, HPC and cloud batch processing, machine- and deep-learning frameworks for image analysis.
- **Geospatial analysis & visualisation:** Generic Mapping Tools (GMT), ArcGIS, Global Mapper, GeoMapApp, SAS.Planet, Google Earth Engine
- **Complementary geophysics:** SEISAN, SAC, Geosoft; GNSS time-series analysis; optical (Sentinel-2 / Landsat) change detection.
- **Reference & writing:** Mendeley, Adobe Illustrator / Canvas for scientific figure production.

TEACHING & MENTORING

Teaching Assistant & Course Tutor — The University of Hong Kong 2022 – 2026
- Introduction to the Atmosphere & Hydrosphere
- Earth Dynamics & Global Tectonics
- Structural Geology (including field and practical sessions)

Teaching Assistant — Institute for Advanced Studies in Basic Sciences, Iran 2016 – 2019
- Programming for Geoscientists
- Basic Mathematics in Geology

Guest Lecturer / Workshop Facilitator 2021 – 2026
- InSAR — theory, processing workflows and geohazard applications (hands-on sessions).
- GeoMapApp — geospatial data exploration, mapping and visualisation.

Certification: Certificate in Teaching and Learning in Higher Education, The University of Hong Kong (2023).

ACADEMIC SERVICE & PROFESSIONAL DEVELOPMENT

- **Peer reviewer,** *Earth and Planetary Science Letters* (EPSL)
- **Professional memberships:** European Geosciences Union (EGU); Seismological Society of America (SSA); Bolin Centre for Climate Research, Stockholm University, Sweden.

Specialist training & workshops

- AMSTer and DefVolc software workshop — Luxembourg, one week, in person (2023).
- InSAR Processing and Analysis with ISCE2 (ISCE+) — one week, online (2022).
- LiCSAR and LiCSBAS training, COMET — University of Leeds, UK, one week, online (2021).
- Pyrocko seismological toolbox — IASBS, Zanjan, Iran, two days, in person (2021).

LANGUAGES

- **Azerbaijani** — native · **Persian** — fluent · **English** — professional working proficiency (IELTS 6.5)
- **Turkish** — intermediate · **Arabic** — basic · **Kurdish** — basic

REFERENCES

Dr. Pablo J. González Ph.D. supervisor
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Associate Professor, Department of Earth Sciences, Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran
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